

Descriptive Analytics Report: Critical Minerals Supply

Global Production, Value, Concentration & Supply-Risk Patterns (2015–2026)

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Tracking Global Production, Market Value, Supply Concentration, Export-Control Exposure & Disruption Risk

Note on data: This report uses the supplied critical_minerals_supply_master dataset containing 1,548 records spanning 2015–2026, covering 24 minerals across 35 countries (129 mineral–country pairs). The dataset is already aggregated at the mineral–country–year level rather than raw transaction-level data. Therefore, the analysis describes the supplied observations and does not assume a larger real-time global minerals data stream.

1. Objective

Build a descriptive analytics layer over the critical minerals supply dataset to answer the following recurring business and policy questions:

- How is global mine production trending over time?
- How is total market value trending over time?
- Which minerals contribute the highest production volume?
- Which minerals contribute the most to overall market value?
- Which countries generate the highest production volumes?
- Which countries contribute the most market value?
- Which minerals are the most geographically concentrated (supply-chain risk)?
- What is the average supply-risk level across the tracked portfolio?
- How are export controls trending, and which minerals/countries are affected?
- How do reserves relate to current production levels?
- How do rare-earth elements differ from other critical minerals in risk and concentration?
- Which minerals have experienced the most supply disruptions?

2. How This Would Work at Real Scale

Monitoring global critical-minerals supply chains is a data-engineering problem before it is an analytics problem. A realistic architecture would ingest production, trade and pricing events, aggregate them by mineral / country / year, retain both raw and processed data, and expose the resulting KPIs through a data-visualization layer.

Layer	Purpose	Typical tools
Investigation	Research, understand and prepare the minerals dataset	Google, Claude & Chat GPT
Stream processing	Aggregate by mineral, country and year	Jupyter Notebook (Python)
Storage	Retain raw and processed data	Excel files / CSV files
Batch aggregation	Annual KPI and summary metrics	Jupyter Notebook
Report	Present analysis, graphs and insights	Data Visualization / MS Word

3. Global Mine Production Trend

Across the supplied 12-year window (2015–2026), the dataset records approximately 936.87 million tonnes of cumulative mine production across all 24 tracked minerals and 35 countries. The chart below shows how annual global production has moved across the available years.

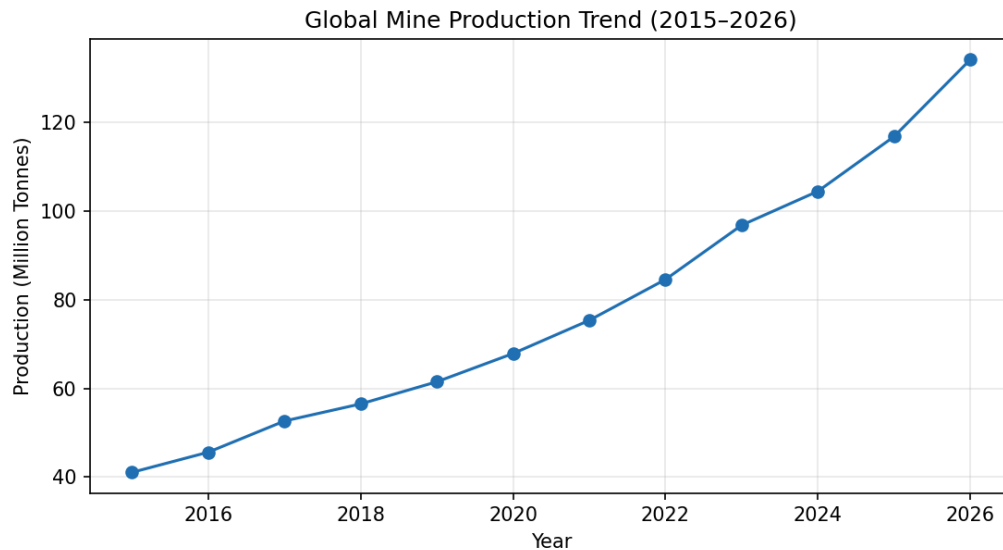


Figure 1. Global mine production trend (2015–2026)

INSIGHTS:

- Global mine production rose from approximately 41.08 million tonnes in 2015 to approximately 134.03 million tonnes in 2026.
- This represents roughly a 3.26x increase over the period, equivalent to an annual growth rate (CAGR) of approximately 11.3%.
- Production growth appears broadly continuous rather than flat, suggesting sustained expansion in mining and extraction activity.
- Monitoring annual production trends can support capacity planning and long-range demand forecasting.

KPI :

Cumulative production (2015–2026): **936.87 Million Tonnes**

2026 annual production: **134.03 Million Tonnes**

Approx. CAGR (2015→2026): **≈11.35% per year**

4. Global Market Value Trend

Total market value in the supplied records (mine production × price per tonne) is approximately \$5.15 trillion, cumulative across all mineral–country–year rows. The average annual value across the represented years is approximately \$428.96 billion, while the highest annual aggregate value is the 2026 figure of approximately \$706.35 billion. Value is calculated directly from the dataset's production and price fields.

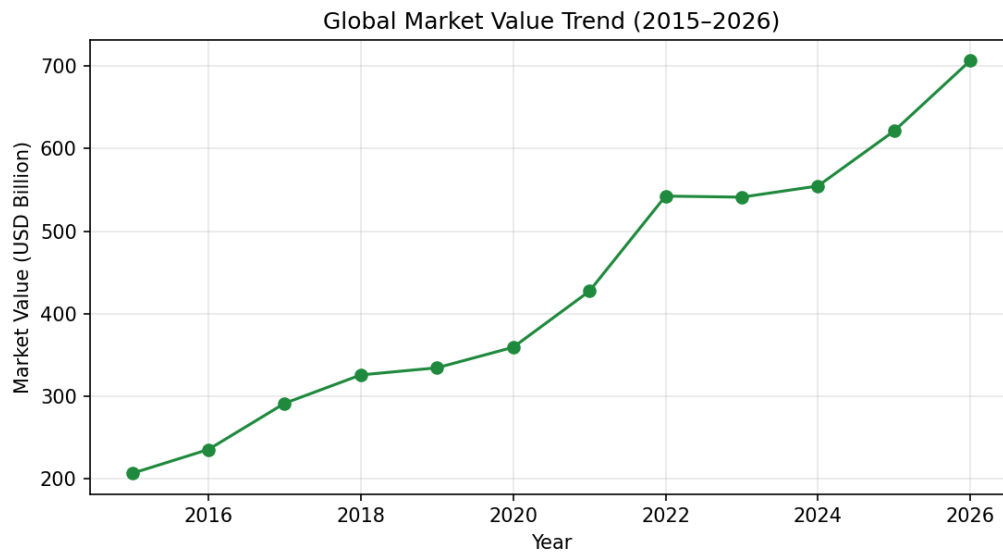


Figure 2. Global market value trend (2015–2026)

INSIGHTS:

- Market value grew from approximately \$206.94 billion in 2015 to approximately \$706.35 billion in 2026 — roughly a 3.41x increase.
- Value has grown slightly faster than tonnage over the period, indicating that price appreciation is compounding the effect of rising output.
- Year-on-year value should be read alongside production volume to separate price-driven swings from genuine output growth.
- Tracking value trends supports investment and market-sizing decisions across the minerals portfolio.

KPI :

Cumulative market value (2015–2026): **≈\$5.15 Trillion**

Average annual value: **≈\$428.96 Billion**

Peak annual value (2026): **≈\$706.35 Billion**

5. Best-Performing Minerals by Production Volume

The dataset spans 24 minerals. Manganese is the largest contributor by production volume, accounting for approximately 52.15% of cumulative tonnage, followed by Copper. The chart below shows the production share for the top 8 minerals, with the remaining 16 minerals grouped as “Other.”

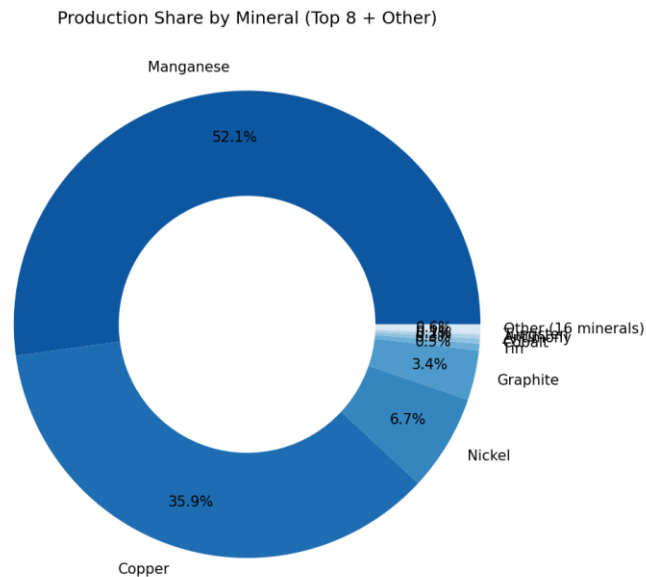


Figure 3. Production share by mineral (top 8 + other)

INSIGHTS:

- Bulk industrial minerals (Manganese, Copper, Nickel) dominate tonnage, together accounting for the large majority of total production volume.
- High-value technology minerals such as rare-earth elements contribute comparatively little tonnage despite their strategic importance.
- Production-volume ranking should not be read as a ranking of economic or strategic importance — see Section 6 for the value-based view.
- Concentration in a small number of minerals by volume can mask risk building in lower-volume, higher-criticality minerals.

KPI :

Top mineral by volume: **Manganese**

Production share: **≈52.15%**

Minerals tracked: **24**

Top 8 Minerals by Production Volume

Mineral	Production (tonnes)	Value (USD)	Volume Share
Manganese	488,549,161.1	\$1,067.70 B	52.15%

Copper	336,779,366.9	\$2,428.14 B	35.95%
Nickel	62,497,464.6	\$909.83 B	6.67%
Graphite	32,300,777.6	\$43.10 B	3.45%
Tin	4,340,648.1	\$92.79 B	0.46%
Cobalt	3,267,725.6	\$119.00 B	0.35%
Antimony	2,262,404.2	\$28.51 B	0.24%
Tungsten	1,299,665.3	\$45.22 B	0.14%

6. Minerals Contributing Most to Overall Value

While Manganese leads by tonnage, Copper is the largest contributor to overall market value, at approximately \$2.428 trillion, or roughly 47.17% of total value. This divergence between the volume leader and the value leader reflects the much higher price per tonne of Copper relative to Manganese.

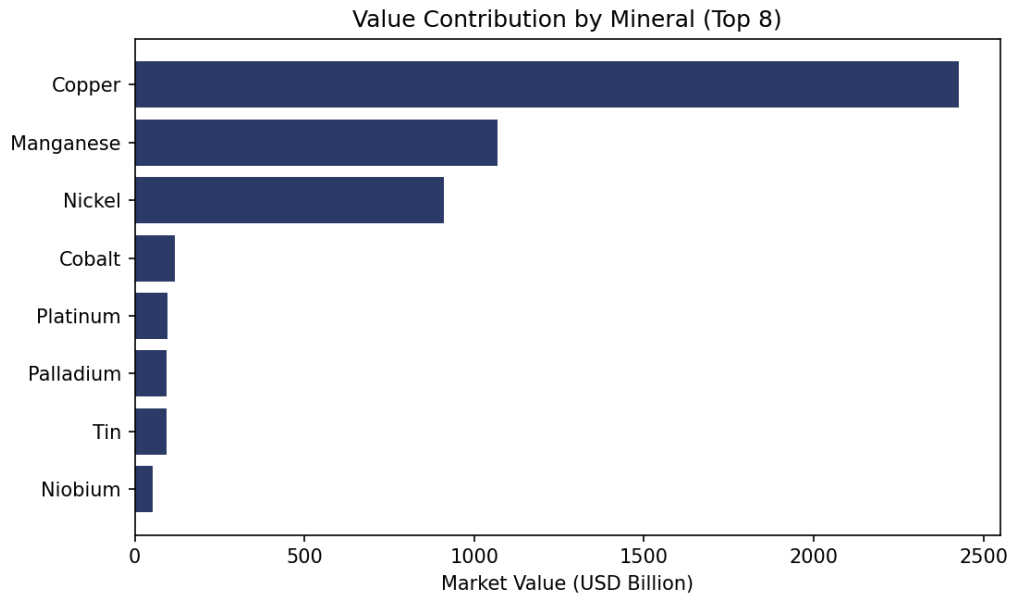


Figure 4. Value contribution by mineral (top 8)

INSIGHTS:

- Copper's high per-tonne price means it drives value even though Manganese produces more tonnage overall.
- Nickel is the third-largest value contributor, reflecting its role in battery-grade supply chains.
- Value concentration in a small set of minerals suggests price shocks in Copper or Manganese would have outsized effects on total market value.
- Volume-based and value-based rankings should both be reviewed together for a complete demand picture.

KPI :

Top mineral by value: **Copper**

Value contribution: **≈\$2.43 Trillion**

Value share: **≈47.17%**

7. Countries That Generate the Highest Production Volumes

South Africa is the largest country-level contributor by production volume, with approximately 166.59 million tonnes ($\approx 17.78\%$ of all supplied production). The distribution shows that production is concentrated in a subset of countries rather than evenly spread across the 35 tracked markets.

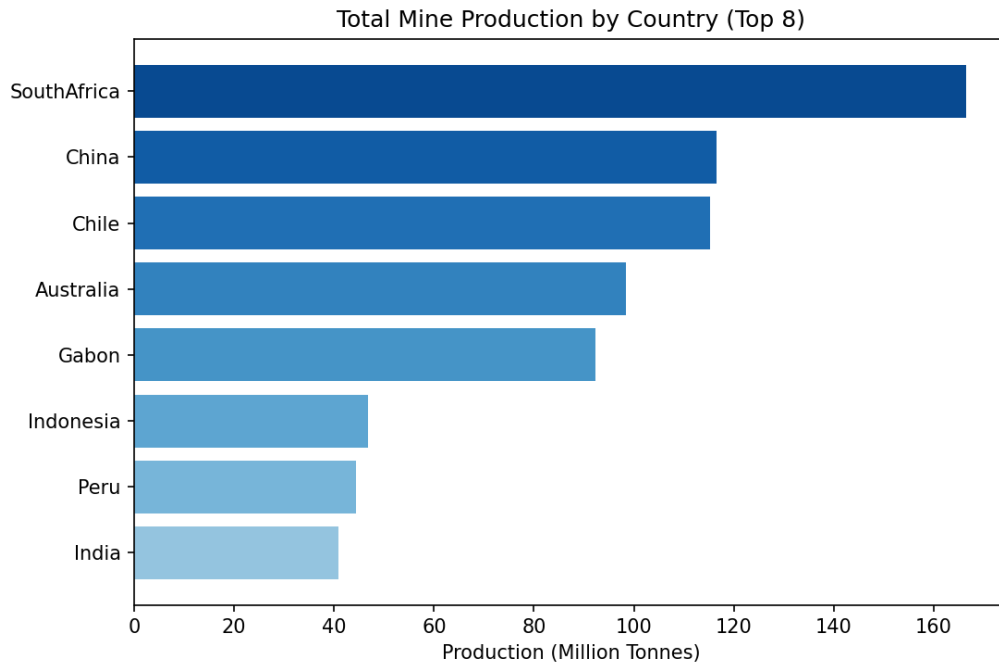


Figure 5. Total mine production by country (top 8)

INSIGHTS:

- Production demand is concentrated in a subset of countries.
- South Africa's lead is driven substantially by Manganese output.
- China, Chile, Australia and Gabon round out the top five, spanning both bulk and battery/technology minerals.
- Country-level production analysis can support sourcing diversification and supplier-risk planning.

KPI :

Top-performing country: **South Africa**

Recorded production: **166.59 Million Tonnes**

Share of total production: **$\approx 17.78\%$**

Top 8 Countries by Production Volume

Country	Production (tonnes)	Value (USD)	Volume Share
South Africa	166,586,890.3	\$475.14 B	17.78%
China	116,591,638.3	\$621.94 B	12.44%
Chile	115,253,627.6	\$837.10 B	12.30%

Australia	98,454,188.7	\$384.57 B	10.51%
Gabon	92,371,031.4	\$201.78 B	9.86%
Indonesia	46,751,914.2	\$523.71 B	4.99%
Peru	44,445,161.3	\$324.70 B	4.74%
India	40,897,294.3	\$88.28 B	4.37%

8. Countries That Contribute the Most Value

Chile is the largest contributor by market value, at approximately \$837.10 billion (≈16.26% of total value), even though it ranks third by tonnage. This is driven by Chile's Copper-heavy production mix, reinforcing the value/volume divergence seen in Section 6.

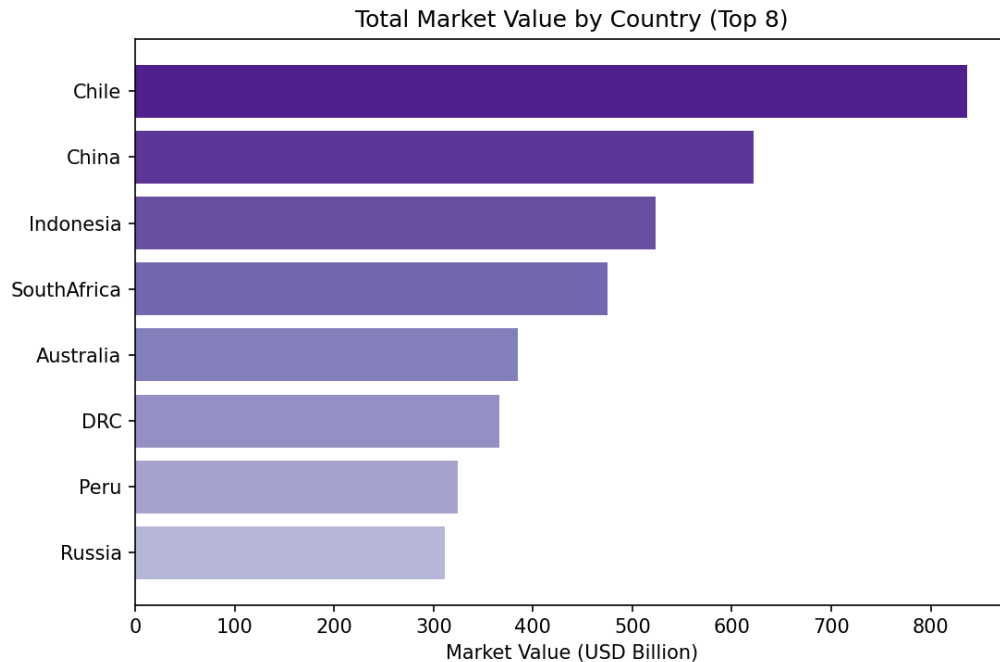


Figure 6. Total market value by country (top 8)

INSIGHTS:

- Value leadership does not always match volume leadership — Chile leads on value while South Africa leads on tonnage.
- China ranks second in both volume and value, reflecting its broad footprint across many mineral categories.
- High-value countries can be important markets for trade-policy and investment planning.
- Country-level value analysis helps identify commercially significant supply relationships.

KPI :

Top-performing country (value): **Chile**

Recorded value: **≈\$837.10 Billion**

Share of total value: **≈16.26%**

9. Most Geographically Concentrated Minerals

Beyond overall country totals, individual minerals vary widely in how concentrated their supply is in a single leading country. Averaged across all tracked years, Gallium is the most concentrated mineral, with a single leading country supplying approximately 94.85% of production. The overall average leading-country share across all minerals is approximately 58.35%, and the average Herfindahl-Hirschman Index (HHI) across the dataset is approximately 0.431.

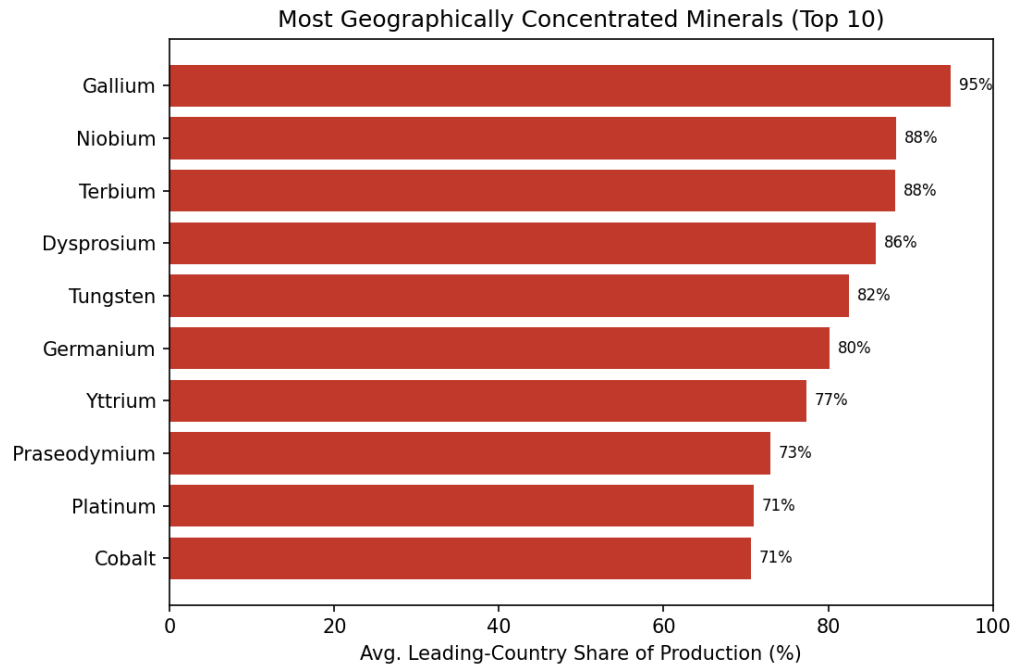


Figure 7. Most geographically concentrated minerals (top 10, by average leading-country share)

INSIGHTS:

- Gallium, Niobium, Terbium and Dysprosium show the highest concentration, each with over 85% average share held by a single leading country.
- Highly concentrated minerals carry elevated single-point-of-failure risk for downstream industries such as semiconductors, magnets and defense manufacturing.
- A high HHI or leading-country share indicates limited near-term substitution options if the leading supplier restricts exports.
- Concentration metrics should be read alongside the Supply Risk Score in Section 10 for a fuller risk picture.

KPI :

Most concentrated mineral: **Gallium**

Avg. leading-country share (Gallium): **≈94.85%**

Avg. leading-country share (all minerals): **≈58.35%**

Avg. HHI (all minerals): **≈0.431**

Top 10 Most Concentrated Minerals

Mineral	Avg. HHI	Avg. Leading-Country Share	Avg. Supply Risk Score
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Gallium	0.903	94.8%	77.6
Niobium	0.799	88.3%	63.4
Terbium	0.769	88.2%	68.1
Dysprosium	0.732	85.8%	67.9
Tungsten	0.682	82.5%	59.1
Germanium	0.650	80.2%	56.9
Yttrium	0.626	77.3%	55.1
Praseodymium	0.543	72.9%	49.2
Platinum	0.537	70.9%	44.8
Neodymium	0.516	70.3%	48.5

10. Supply Risk Score

The average supply risk score across the supplied records is 37.67 (on a 0–100 scale, where higher indicates greater risk).

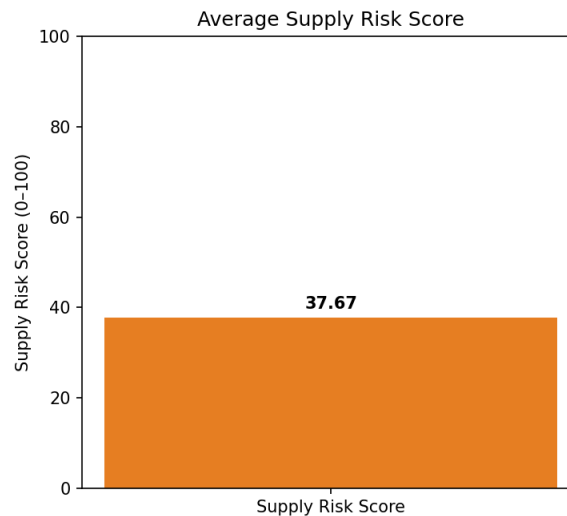


Figure 8. Average supply risk score

INSIGHTS:

- Mineral-level differences: a mineral-by-mineral breakdown can reveal which categories carry materially higher risk than the portfolio average.
- Approximately 30.04% of all records (465 of 1,548) are flagged as high supply risk, indicating a meaningful minority of mineral–country combinations warrant closer monitoring.
- Supply risk score should be monitored continuously, since changes may signal emerging operational or geopolitical issues.
- The 2026 average risk score (≈ 37.90) is close to the 12-year average, suggesting risk levels have not materially eased in the most recent year.

KPI :

Average supply risk score: **37.67 / 100**

High-supply-risk records: **465 ($\approx 30.04\%$)**

2026 average risk score: **37.90 / 100**

11. Export Control & Regional Risk Exposure

Export-control activity is heavily concentrated in the most recent years of the dataset. Across all 1,548 records, only 25 ($\approx 1.61\%$) show an active export control, and all of these appear from 2023 onward — none are recorded between 2015 and 2022.

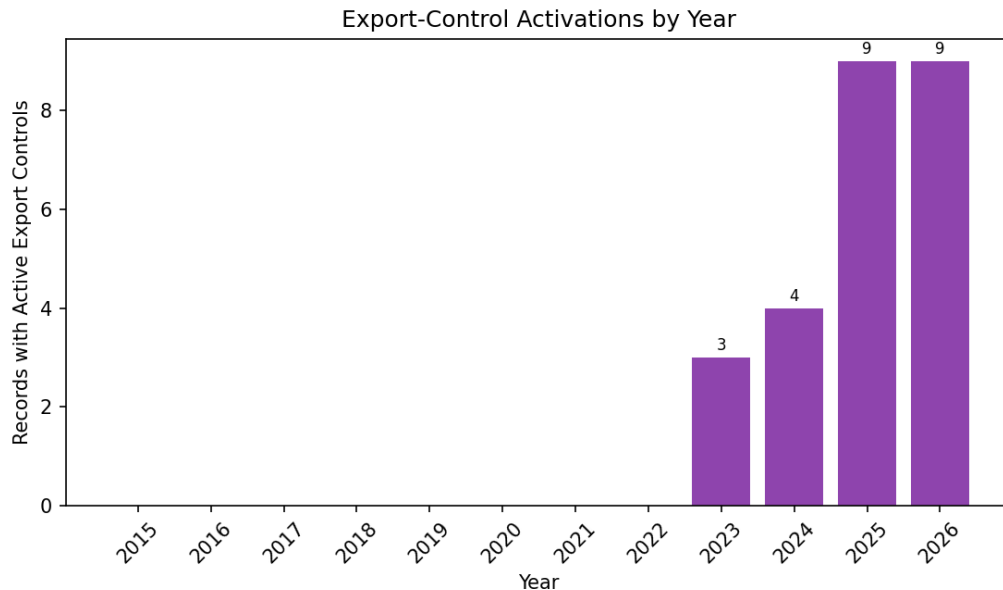


Figure 9. Export-control activations by year

INSIGHTS:

- Export-control activations begin in 2023 (3 records) and rise to 9 records each in 2025 and 2026, indicating an escalating trend within the supplied window.
- China accounts for 23 of the 25 export-control-flagged records, with DRC accounting for the remaining 2.
- Gallium, Germanium and Graphite are the most frequently export-control-flagged minerals (4 records each), followed by Antimony (3).
- These minerals overlap substantially with the most geographically concentrated minerals identified in Section 9, reinforcing the link between concentration and export-policy exposure.

KPI :

Export-control-active records: **25 ($\approx 1.61\%$)**

Years affected: **2023 – 2026**

Leading flagged country: **China (23 of 25 records)**

12. Reserves vs. Production Relationship

The relationship between reserves and current production provides an additional descriptive view of supply sustainability. Mineral–country pairs with larger reserve bases tend to have proportionally larger current production, but the scatter plot should be interpreted as association rather than proof of causation.

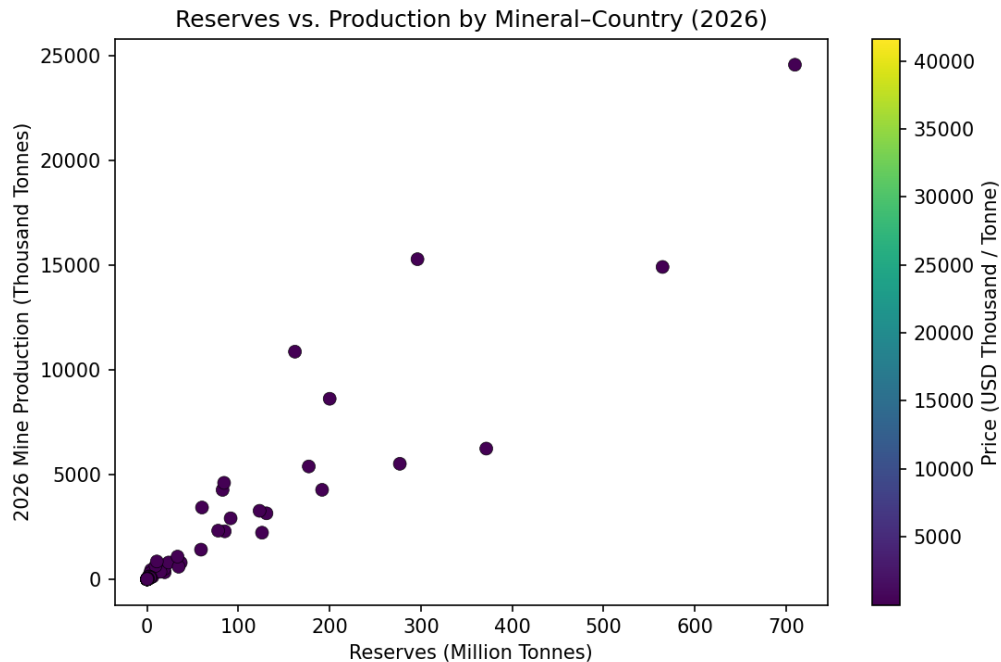


Figure 10. Reserves vs. production by mineral–country (2026)

INSIGHTS:

- Reserves and production show a strong positive correlation (≈ 0.90) across the dataset, indicating that larger reserve bases are generally associated with larger current output.
- The average years-of-reserves figure across all records is approximately 34.15 years, implying a moderate long-run production runway at current extraction rates.
- Rare-earth mineral–country pairs show a slightly shorter average reserve runway (≈ 32.85 years) than non-rare-earth pairs (≈ 34.56 years).
- Higher-priced minerals (shown by color) do not consistently sit at the high-reserve, high-production end of the chart, suggesting price is driven by more than raw resource abundance.

KPI :

Reserves–production correlation: **≈ 0.90**

Avg. years of reserves (all records): **≈ 34.15 years**

Avg. years of reserves (rare-earth only): **≈ 32.85 years**

13. Rare-Earth vs. Non-Rare-Earth Production Patterns

Rare-earth element (REE) records behave very differently from non-REE records in the dataset. REEs average approximately 6,707 tonnes of production per record versus approximately 794,539 tonnes for non-REEs, but carry substantially higher average supply risk and concentration scores.

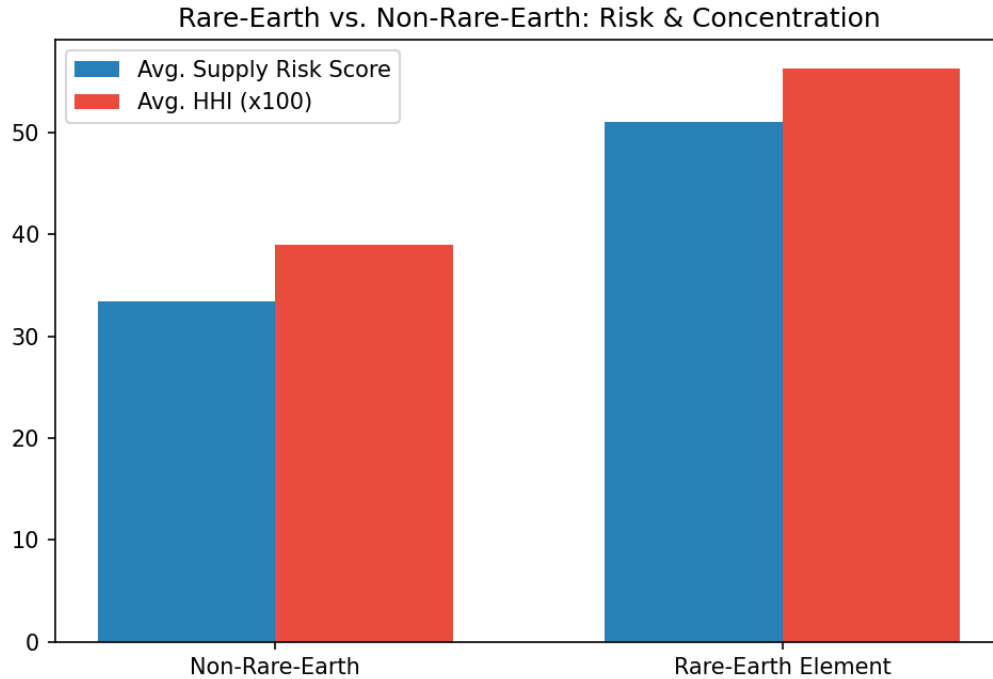


Figure 11. Rare-earth vs. non-rare-earth: risk & concentration

INSIGHTS:

- The average supply risk score for rare-earth records is 51.06, compared to 33.43 for non-rare-earth records — a substantial gap.
- Average HHI is also higher for rare-earth records (≈ 0.563) than for non-rare-earth records (≈ 0.390), confirming greater supply-side concentration.
- Despite far lower production tonnage, rare-earth minerals carry disproportionate strategic weight because of their role in magnets, semiconductors and other high-technology end uses.
- This pattern should be treated as a descriptive comparison of the available observations, not a causal claim about why rare earths carry more risk.

KPI :

Avg. supply risk — Rare-earth: **51.06 / 100**

Avg. supply risk — Non-rare-earth: **33.43 / 100**

Rare-earth records: **372 ($\approx 24.03\%$ of dataset)**

14. Supply Disruption Frequency

Across the supplied records, 71 ($\approx 4.59\%$) are flagged with a supply disruption event. Lithium has the highest disruption count among individual minerals, followed by Cobalt and Nickel — all central to battery supply chains.

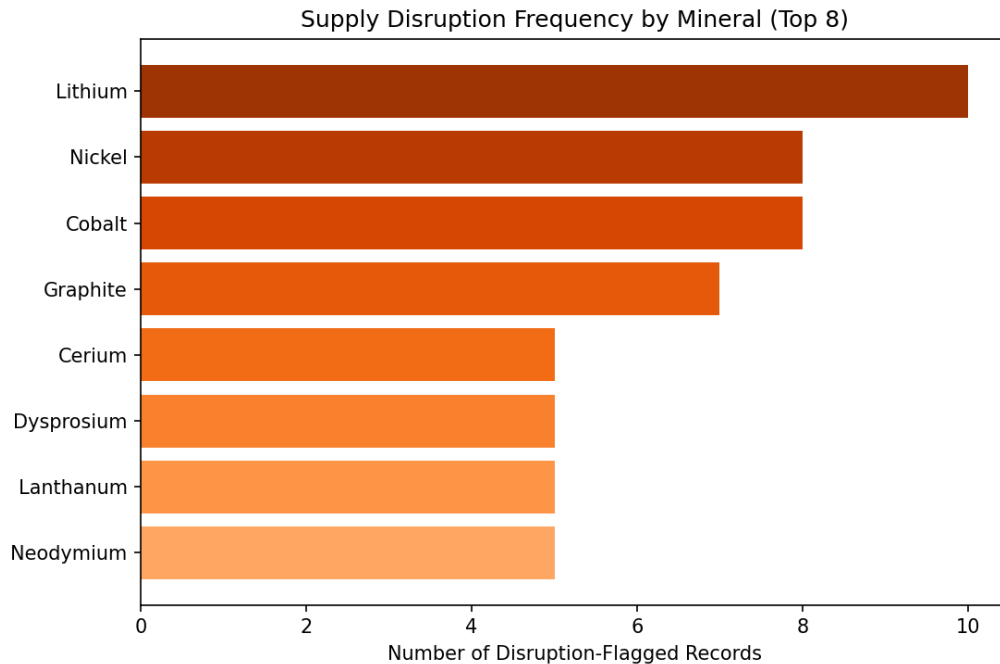


Figure 12. Supply disruption frequency by mineral (top 8)

INSIGHTS:

- Disruption events are heavily concentrated in a single year — 2021 accounts for 57 of the 71 flagged records — with much lower counts in surrounding years.
- Battery-chain minerals (Lithium, Cobalt, Nickel, Graphite) account for the majority of disruption-flagged records among individual minerals.
- China (20 records), Australia (11) and Russia (8) are the countries most frequently associated with disruption flags.
- Disruption frequency is a useful operational-reliability signal alongside the Supply Risk Score in Section 10 and the export-control trend in Section 11.

KPI :

Disruption-flagged records: **71 ($\approx 4.59\%$)**

Peak disruption year: **2021 (57 events)**

Most disruption-prone mineral: **Lithium (10 events)**

15. Production by End-Use Category

The dataset also classifies each mineral by its primary industrial end use. Battery-related minerals account for the largest share of production volume ($\approx 62.71\%$), followed by grid/electrical applications ($\approx 35.95\%$). The remaining nine end-use categories together account for less than 2% of total tonnage, though several — such as magnets and semiconductors — carry outsized strategic and technological importance relative to their volume.

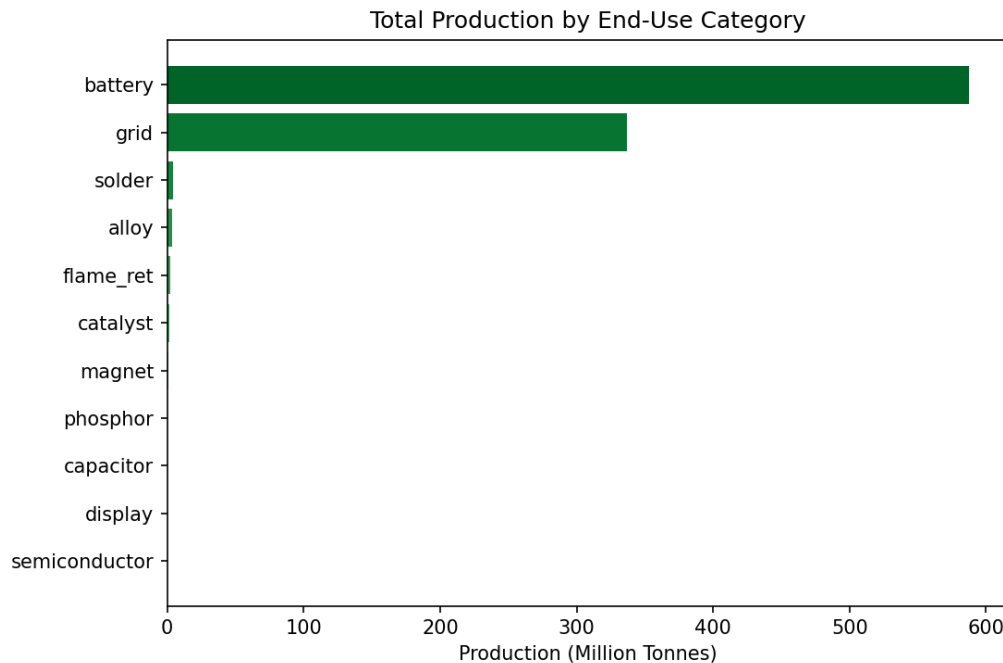


Figure 13. Total production by end-use category

INSIGHTS:

- Battery and grid applications together account for roughly 98.7% of tracked production volume, reflecting the dataset's weighting toward energy-transition minerals.
- Magnet, semiconductor, capacitor and display end-uses represent a very small share of tonnage but draw heavily on the most concentrated and highest-risk minerals identified in Sections 9 and 13.
- Grid-related production carries the highest average value per tonne among the major end-use categories, consistent with Copper's large weighting in that category.
- End-use segmentation helps connect upstream mineral-supply risk to specific downstream industries (e.g., EV batteries vs. power infrastructure vs. electronics).

KPI :

Largest end-use category: **Battery ($\approx 62.71\%$ of production)**

Second-largest end-use category: **Grid ($\approx 35.95\%$ of production)**

End-use categories tracked: **11**

Summary of Key Metrics

Metric	Value
Total records	1,548
Years covered	2015 – 2026 (12 years)
Minerals tracked	24
Countries tracked	35
Mineral–country pairs	129
Cumulative production	936.87 Million Tonnes
Cumulative market value	≈\$5.15 Trillion
2026 production	134.03 Million Tonnes
2026 market value	≈\$706.35 Billion
Average supply risk score	37.67 / 100
High-supply-risk records	465 (≈30.04%)
Export-control-active records	25 (≈1.61%)
Disruption-flagged records	71 (≈4.59%)
Rare-earth records	372 (≈24.03%)

Next Steps to Move from Descriptive → Predictive

- Demand forecasting by mineral and country to estimate future production and value trajectories.
- Anomaly detection for sudden changes in supply risk score, price or export-control status.
- Export-control impact analysis linking control activations to subsequent price and disruption outcomes.
- Reserve-depletion modeling using years-of-reserves trends to flag minerals approaching supply constraints.
- Scenario and stress testing for highly concentrated minerals, simulating the effect of a leading-country supply interruption.

This report answers what happened in the supplied dataset. The next analytical layer would use longer historical coverage and richer event-level trade data to explain why changes occurred and to forecast what is likely to happen next.

Glossary

I. HHI — Herfindahl-Hirschman Index: A standard measure of market concentration calculated as the sum of squared market shares of all suppliers, ranging from near 0 (highly fragmented) to 1 (single supplier monopoly).

II. Supply Risk Score: A composite 0–100 indicator in the dataset reflecting the overall vulnerability of a mineral–country supply line; higher values indicate greater risk.

III. Rare Earth Element (REE): One of a group of 17 chemically similar elements (including minerals such as Cerium, Neodymium, Dysprosium and Terbium) used mainly in magnets, catalysts and electronics, and typically produced in far smaller volumes than bulk industrial metals.

IV. Export Control: A government-imposed restriction on the export of a mineral or related product, which can limit supply availability to importing countries.

V. Years of Reserves: An estimate of how many years current reserves would last at the present rate of production; a rough, simplified sustainability indicator rather than a precise forecast.

VI. Refined Share: The percentage of a country's production that is processed into refined (rather than raw/ore-stage) material, relevant to downstream supply-chain capacity.

VII. Disruption Flag: A binary indicator in the dataset marking whether a mineral–country–year record experienced a recorded supply disruption event.